Reduced Order Waveform Relaxation

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Abstract

aim is to show things work as predicted...

1 Introduction

We consider the heat equation on a bounded domain $\Omega \subset \mathbb{R}^n$ with a smooth boundary $\partial\Omega$

$$
\begin{aligned}
\frac{\partial u}{\partial t} &= \Delta u + f(x,t), && x \in \Omega, && 0 < t < T, \\
u(x,t) &= g(x,t), && x \in \partial\Omega, && 0 < t < T, \\
u(x,0) &= u_0(x), && x \in \Omega,
\end{aligned}
$$

where the initial condition $u_0(x)$ and the boundary condition $g(x,t)$ are bounded piecewise continuous and $f(x,t)$ is continuous. On each subdomain Ω_j, we solve at each step $k+1$ of the waveform relaxation iteration the subproblem

$$
\begin{aligned}
\frac{\partial u_j^{k+1}}{\partial t} &= \Delta u_j^{k+1} + f(x,t), \\
u_j^{k+1}(x,t) &= u_l^k(x,t), && x \in \Gamma_{jl}, && 0 < t < T, \\
u_j^{k+1}(x,t) &= g(x,t), && x \in \Gamma_{i0}, && 0 < t < T, \\
u_j^{k+1}(x,0) &= u_0(x), && x \in \Omega_j,
\end{aligned}
$$

for $j = 1, \dots, N$. Let us define the transmission operator $\mathcal{T}_{jl} : V_l \to V_l|_{\Gamma_{jl}}$ by

$$
\mathcal{T}_{jl} u_l^k := u_l^k \Big|_{\Gamma_{jl}},
$$

where $u_l^k \in V_l$ is the solution on subdomain l at iteration k, V_l denotes the function space associated with subdomain l, and Γ_{jl} is the interface between subdomains j and l. This operator extracts the values of the solution u_l^k on the interface Γ_{jl} between subdomains j and l, which are then used as boundary data for subdomain j in the overlapping waveform relaxation iteration.

Let $V_{j,r} \subset V_j$ be the reduced space, constructed from snapshots of u_j or a global solution approximation. On each subdomain Ω_j, we solve at each step $k+1$ of the reduced-order waveform relaxation iteration the subproblem

$$
\begin{aligned}
\left(\partial_t u_{j,r}^{k+1}, v_r\right)_{\Omega_j} &= \left(\Delta u_{j,r}^{k+1}, v_r\right)_{\Omega_j} + (f, v_r)_{\Omega_j}, && \forall v_r \in V_{j,r}, \\
u_{j,r}^{k+1}(x,t) &= u_{l,r}^k(x,t), && x \in \Gamma_{jl}, \quad 0 < t < T, \\
u_{j,r}^{k+1}(x,t) &= P_{j,r} g(x,t), && x \in \Gamma_{j0}, \quad 0 < t < T, \\
u_{j,r}^{k+1}(x,0) &= P_{j,r} u_0(x), && x \in \Omega_j,
\end{aligned}
$$

where the projection operator $P_{j,r} : V_j \to V_{j,r}$ is defined as

$$P_{j,r}u := \arg\min_{v_r \in V_{j,r}} \|u - v_r\|_{V_j},$$

i.e., for any function $u \in V_j$, $P_{j,r}u$ is the best approximation of u in the reduced space $V_{j,r}$ with respect to the norm $\|\cdot\|_{V_j}$. We note that the transmission operator defined above can also be applied to reduced-order solutions $u_{l,r}^k \in V_{l,r} \subset V_l$, so that the reduced interface condition in the subproblem can be written compactly as

$$u_{j,r}^{k+1}\Big|_{\Gamma_{jl}} = \mathcal{T}_{jl} u_{l,r}^k.$$

Now, let us define the error

$$e_{j,r}^{k+1} := u_j^{k+1} - u_{j,r}^{k+1} \in V_j.$$

We can split this error as

$$e_{j,r}^{k+1} = \left(u_j^{k+1} - P_{j,r}u_j^{k+1}\right) + \left(P_{j,r}u_j^{k+1} - u_{j,r}^{k+1}\right).$$

Let us introduce the following

$$\eta_j^{k+1} = u_j^{k+1} - P_{j,r}u_j^{k+1}, \qquad \mu_j^{k+1} = P_{j,r}u_j^{k+1} - u_{j,r}^{k+1},$$

then the error becomes

$$e_{j,r}^{k+1} = \eta_j^{k+1} + \mu_j^{k+1}.$$

In the original error $e_{j,r}^{k+1}$, we are comparing functions in different spaces. That's why we split in such a way so that we can define error in a compatible way.

Let us concentrate on the error μ_j^{k+1}. It is clear that $P_{j,r}u_j^{k+1} \in V_{j,r}$. Then the projected problem reads

$$\left(\partial_t P_{j,r}u_j^{k+1}, v_r\right)_{\Omega_j} = \left(\Delta P_{j,r}u_j^{k+1}, v_r\right)_{\Omega_j} + (f, v_r)_{\Omega_j} - \left(\Delta\left(P_{j,r}u_j^{k+1} - u_j^{k+1}\right), v_r\right)_{\Omega_j}, \quad \forall v_r \in V_{j,r},$$

where $\Delta\left(P_{j,r}u_j^{k+1} - u_j^{k+1}\right)$ represents the residual caused by the projection. Subtracting the reduced-order waveform relaxation equation from this equation, we obtain

$$\left(\partial_t P_{j,r}u_j^{k+1} - \partial_t u_{j,r}^{k+1}, v_r\right)_{\Omega_j} = \left(\Delta\left(P_{j,r}u_j^{k+1} - u_{j,r}^{k+1}\right), v_r\right)_{\Omega_j} - \left(\Delta\left(P_{j,r}u_j^{k+1} - u_j^{k+1}\right), v_r\right)_{\Omega_j}, \forall v_r \in V_{j,r}.$$

Considering the defined error definitions above, this reduces to

$$\left(\partial_t \mu_j^{k+1}, v_r\right)_{\Omega_j} = \left(\Delta\mu_j^{k+1}, v_r\right)_{\Omega_j} + \left(\Delta\eta_j^{k+1}, v_r\right)_{\Omega_j}, \qquad \forall v_r \in V_{j,r}.$$

Now, let us consider transmission conditions for μ_j^{k+1}. Since

$$u_j^{k+1}\Big|_{\Gamma_{jl}} = \mathcal{T}_{jl}u_l^k, \qquad u_{j,r}^{k+1}\Big|_{\Gamma_{jl}} = \mathcal{T}_{jl}u_{l,r}^k,$$

on the interface boundary Γ_{jl}, we have

$$\begin{aligned}
\mu_j^{k+1}\Big|_{\Gamma_{jl}} &= \left(P_{j,r}u_j^{k+1} - u_{j,r}^{k+1}\right)\Big|_{\Gamma_{jl}} = P_{j,r}\mathcal{T}_{jl}u_l^k - \mathcal{T}_{jl}u_{l,r}^k = \mathcal{T}_{jl}\mu_l^k + P_{j,r}\mathcal{T}_{jl}u_l^k - \mathcal{T}_{jl}P_{l,r}u_l^k = \\
&= \mathcal{T}_{jl}\mu_l^k + \mathcal{T}_{jl}\eta_l^k + P_{j,r}\mathcal{T}_{jl}u_l^k - \mathcal{T}_{jl}u_l^k = \mathcal{T}_{jl}\mu_l^k + \mathcal{T}_{jl}\eta_l^k + (P_{j,r} - I)\,\mathcal{T}_{jl}u_l^k = \\
&= \mathcal{T}_{jl}\left(\mu_l^k + \eta_l^k\right) + (P_{j,r} - I)\,\mathcal{T}_{jl}u_l^k = \mathcal{T}_{jl}e_{l,r}^k + (P_{j,r} - I)\,\mathcal{T}_{jl}u_l^k.
\end{aligned}$$

Remark 1. The equality

$$\mu_j^{k+1}\Big|_{\Gamma_{jl}} = \mathcal{T}_{jl} e_{l,r}^k + (P_{j,r} - I)\, \mathcal{T}_{jl} u_l^k$$

expresses how the interface error in the reduced problem is formed. One part of the interface error is simply the neighboring subdomain's total error transported through the transmission operator, which is exactly the same mechanism by which errors propagate in the full waveform relaxation method and has nothing to do with model reduction. The other part is a purely reduction-induced defect: even if the neighboring solution were exact, projecting the transmitted interface data into the reduced space generally introduces a mismatch because the reduced space cannot represent all possible traces. Our reduced space $V_{j,r}$ is built from a finite number of snapshots of solutions inside the subdomain. These snapshots capture the dominant interior behavior of the solution, but they do not span all functions that can appear on the interface Γ_{jl}. In particular, the interface data coming from a neighboring subdomain through the transmission operator may contain spatial or temporal features that were never present in the snapshot set. When such interface data is imposed on the reduced subproblem, it must first be projected onto $V_{j,r}$, and during this projection, any component that lies outside the span of the reduced basis is inevitably lost. *This loss shows up as a mismatch between the exact transmitted interface value and its reduced representation.* Therefore, even if the neighboring solution were exact, the reduced subdomain cannot reproduce its trace perfectly unless the reduced space is rich enough to contain that trace. This is precisely what is meant by saying that the reduced space cannot represent all possible traces, and it is why the additional projection term appears in the interface condition for the reduced error. Thus, the interface value of the reduced error is the sum of the propagated waveform relaxation error and a fixed consistency error caused by the projection. This separation is fundamental, because it shows that the iterative behavior of the reduced method mirrors that of the full method, while the influence of model reduction enters only through an additional, clearly identifiable projection error term.

On the interface boundary Γ_{j0}, we have

$$\mu_j^{k+1}\Big|_{\Gamma_{j0}} = \left(P_{j,r} u_j^{k+1} - u_{j,r}^{k+1}\right)\Big|_{\Gamma_{j0}} = P_{j,r} g(x,t) - P_{j,r} g(x,t) = 0.$$

The initial condition becomes

$$\mu_j^{k+1}(x,0) = 0.$$

Therefore, the problem for $\mu_j^{k+1} \in V_{j,r}$ is found as

$$\begin{aligned}
\left(\partial_t \mu_j^{k+1}, v_r\right)_{\Omega_j} &= \left(\Delta \mu_j^{k+1}, v_r\right)_{\Omega_j} + \left(\Delta \eta_j^{k+1}, v_r\right)_{\Omega_j}, && \forall v_r \in V_{j,r},\\
\mu_j^{k+1}(x,t) &= \mathcal{T}_{jl} e_{l,r}^k + (P_{j,r} - I)\, \mathcal{T}_{jl} u_l^k, && x \in \Gamma_{jl}, \quad 0 < t < T,\\
\mu_j^{k+1}(x,t) &= 0, && x \in \Gamma_{j0}, \quad 0 < t < T,\\
\mu_j^{k+1}(x,0) &= 0, && x \in \Omega_j.
\end{aligned}$$

Remark 2. As the reduced space $V_{j,r}$ is enriched, the projection operator $P_{j,r}$ converges to the identity on V_j. As a result, the projection error $\eta_j^{k+1} = u_j^{k+1} - P_{j,r} u_j^{k+1}$ diminishes, and the additional interface discrepancy $(P_{j,r} - I)\mathcal{T}_{jl} u_l^k$ becomes negligible. In this limit, the reduced error component μ_j^{k+1} increasingly coincides with the total error $e_{j,r}^{k+1}$, which is already evident from the decomposition $e_{j,r}^{k+1} = \eta_j^{k+1} + \mu_j^{k+1}$. Consequently, for sufficiently rich reduced spaces, the reduced-order waveform relaxation method reproduces the error propagation behavior of the full method, with the influence of model reduction becoming asymptotically negligible.

Let us now choose

$$v_r = \mu_j^{k+1} \in V_{j,r}.$$

Then, the left-hand side becomes

$$\left(\partial_t \mu_j^{k+1}, \mu_j^{k+1}\right)_{\Omega_j} = \frac{1}{2}\frac{d}{dt}\left(\mu_j^{k+1}, \mu_j^{k+1}\right)_{\Omega_j} = \frac{1}{2}\frac{d}{dt}\left\|\mu_j^{k+1}\right\|^2_{L^2(\Omega_j)}.$$

For the first term on the left-hand side, using Green's first identity (integration by parts), we obtain

$$\begin{aligned}\left(\Delta\mu_j^{k+1},\mu_j^{k+1}\right)_{\Omega_j} &= \int_{\Omega_j}\Delta\mu_j^{k+1}\mu_j^{k+1} = -\int_{\Omega_j}\nabla\mu_j^{k+1}\cdot\nabla\mu_j^{k+1} + \int_{\partial\Omega_j}\mu_j^{k+1}\nabla\mu_j^{k+1}\cdot n = \\ &= -\int_{\Omega_j}\nabla\mu_j^{k+1}\cdot\nabla\mu_j^{k+1} + \sum_{l\in\mathcal{N}_j}\int_{\Gamma_{jl}}\mu_j^{k+1}\nabla\mu_j^{k+1}\cdot n = \\ &= -\|\nabla\mu_j^{k+1}\|^2_{L^2(\Omega_j)} + \sum_{l\in\mathcal{N}_j}\int_{\Gamma_{jl}}\mu_j^{k+1}\nabla\mu_j^{k+1}\cdot n,\end{aligned}$$

where the neighbor index set is defined as

$$\mathcal{N}_i = \{j\in\{1,\dots,N\} : (\partial\Omega_i\cap\Omega_j)\setminus\partial\Omega\neq\emptyset\}.$$

Since $\mu_j^{k+1}\Big|_{\Gamma_{jl}} = \mathcal{T}_{jl}e_{l,r}^k + (P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k$, the above becomes

$$\left(\Delta\mu_j^{k+1},\mu_j^{k+1}\right)_{\Omega_j} = -\|\nabla\mu_j^{k+1}\|^2_{L^2(\Omega_j)} + \sum_{l\in\mathcal{N}_j}\int_{\Gamma_{jl}}\left(\mathcal{T}_{jl}e_{l,r}^k + (P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\right)\nabla\mu_j^{k+1}\cdot n.$$

Thus, the general equation becomes

$$\frac{1}{2}\frac{d}{dt}\left\|\mu_j^{k+1}\right\|^2_{L^2(\Omega_j)} + \|\nabla\mu_j^{k+1}\|^2_{L^2(\Omega_j)} = \sum_{l\in\mathcal{N}_j}\int_{\Gamma_{jl}}\left(\mathcal{T}_{jl}e_{l,r}^k + (P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\right)\nabla\mu_j^{k+1}\cdot n + \left(\Delta\eta_j^{k+1},v_r\right)_{\Omega_j}.$$

Since $\mu_j^{k+1}\in V_{j,r}\subset V_j\subset H^1(\Omega_j)$, then its trace $\mu_j^{k+1}\Big|_{\partial\Omega_j}\in H^{1/2}(\partial\Omega_j)$, and the (weak) normal derivative $\nabla\mu_j^{k+1}\cdot n$ is well-defined as an element of the dual space $H^{-1/2}(\partial\Omega_j)$, in particular on the interface $\Gamma_{jl}\subset\partial\Omega_j$. In other words, $g_j^{k+1} = \mathcal{T}_{jl}e_{l,r}^k + (P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\in H^{1/2}(\Gamma_{jl})$. Using the Cauchy-Schwarz inequality, we get

$$\left|\int_{\Gamma_{jl}}g_j^{k+1}\nabla\mu_j^{k+1}\cdot n\right| \leq \|\mathcal{T}_{jl}e_{l,r}^k + (P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\|_{H^{1/2}(\Gamma_{jl})}\|\nabla\mu_j^{k+1}\cdot n\|_{H^{-1/2}(\Gamma_{jl})}.$$

Before proceeding, note that g_j^{k+1} comes from subdomain Ω_l and is only defined on the interface. Let us consider the first term on the right-hand side:

$$\|\mathcal{T}_{jl}e_{l,r}^k + (P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\|_{H^{1/2}(\Gamma_{jl})} \leq \|\mathcal{T}_{jl}e_{l,r}^k\|_{H^{1/2}(\Gamma_{jl})} + \|(P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\|_{H^{1/2}(\Gamma_{jl})}.$$

Using the note stated above and the interior trace[1] inequality, we have

$$\|\mathcal{T}_{jl}e_{l,r}^k\|_{H^{1/2}(\Gamma_{jl})} \leq C_{1,l}\|e_{l,r}^k\|_{H^1(\Omega_l)},$$

and the standard trace inequality

$$\|\nabla\mu_j^{k+1}\cdot n\|_{H^{-1/2}(\Gamma_{jl})} \leq C_{2,l}\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}.$$

The interface defect is exactly the distance between the transmitted exact interface data and the reduced trace space

$$\|(P_{j,r}-I)\,\mathcal{T}_{jl}u_l^k\|_{H^{1/2}(\Gamma_{jl})} = \inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})}.$$

Then, we obtain

$$\left|\int_{\Gamma_{jl}}g_j^{k+1}\nabla\mu_j^{k+1}\cdot n\right| \leq C_{2,l}\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}\left(C_{1,l}\|e_{l,r}^k\|_{H^1(\Omega_l)} + \inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})}\right).$$

[1] In a 2-dimensional domain, interior trace means curve inside the domain

Let us focus on $\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}\|e_{l,r}^k\|_{H^1(\Omega_l)}$. Now, let us use Young's inequality to get

$$\|e_{l,r}^k\|_{H^1(\Omega_l)}\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)} \leq \frac{\varepsilon}{2}\|e_{l,r}^k\|_{H^1(\Omega_l)}^2 + \frac{1}{2\varepsilon}\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}^2, \quad \varepsilon > 0,$$
$$\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})}\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)} \leq \frac{\tilde{\varepsilon}}{2}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})}^2 + \frac{1}{2\tilde{\varepsilon}}\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}^2, \quad \tilde{\varepsilon} > 0.$$

The inequality becomes

$$\left|\int_{\Gamma_{jl}} g_j^{k+1}\nabla\mu_j^{k+1}\cdot n\right| \leq C_{1,l}C_{2,l}\frac{\varepsilon}{2}\|e_{l,r}^k\|_{H^1(\Omega_l)}^2 + \left(C_{1,l}C_{2,l}\frac{1}{2\varepsilon} + C_{2,l}\frac{1}{2\tilde{\varepsilon}}\right)\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}^2 + $$
$$+ C_{2,l}\frac{\tilde{\varepsilon}}{2}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})}^2.$$

For the last term on the right-hand side, if we use the Cauchy-Schwarz inequality and Young's inequality, we get

$$\left|\left(\Delta\eta_j^{k+1}, \mu_j^{k+1}\right)_{\Omega_j}\right| \leq \|\Delta\eta_j^{k+1}\|_{L^2(\Omega_j)}\|\mu_j^{k+1}\|_{L^2(\Omega_j)} \leq \frac{\hat{\varepsilon}}{2}\|\Delta\eta_j^{k+1}\|_{L^2(\Omega_j)}^2 + \frac{1}{2\hat{\varepsilon}}\|\mu_j^{k+1}\|_{L^2(\Omega_j)}^2, \quad \hat{\varepsilon} > 0.$$

Considering all the inequalities derived for the right-hand side terms of the equation, we have

$$\frac{1}{2}\frac{d}{dt}\left\|\mu_j^{k+1}\right\|_{L^2(\Omega_j)}^2 + \|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}^2 \leq \frac{\varepsilon}{2}\sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e_{l,r}^k\|_{H^1(\Omega_l)}^2 + \frac{\tilde{\varepsilon}}{2}\sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})} +$$
$$+ \sum_{l\in\mathcal{N}_j}\left(C_{1,l}C_{2,l}\frac{1}{2\varepsilon} + C_{2,l}\frac{1}{2\tilde{\varepsilon}}\right)\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}^2 +$$
$$+ \frac{\hat{\varepsilon}}{2}\|\Delta\eta_j^{k+1}\|_{L^2(\Omega_j)}^2 + \frac{1}{2\hat{\varepsilon}}\|\mu_j^{k+1}\|_{L^2(\Omega_j)}^2,$$

or equivalently,

$$\frac{1}{2}\frac{d}{dt}\left\|\mu_j^{k+1}\right\|_{L^2(\Omega_j)}^2 \leq \frac{\varepsilon}{2}\sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e_{l,r}^k\|_{H^1(\Omega_l)}^2 + \frac{\tilde{\varepsilon}}{2}\sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})} +$$
$$+ \left(\sum_{l\in\mathcal{N}_j}\left(C_{1,l}C_{2,l}\frac{1}{2\varepsilon} + C_{2,l}\frac{1}{2\tilde{\varepsilon}}\right) - 1\right)\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)}^2 +$$
$$+ \frac{\hat{\varepsilon}}{2}\|\Delta\eta_j^{k+1}\|_{L^2(\Omega_j)}^2 + \frac{1}{2\hat{\varepsilon}}\|\mu_j^{k+1}\|_{L^2(\Omega_j)}^2.$$

Recall that the inverse estimate for finite-element space (see 4.5 Inverse Estimates in "The Mathematical Theory of Finite Element Methods" by Susanne C. Brenner and L. Ridgway Scott) is given by

$$\|v\|_{H^1(\Omega)} = \left(\sum_{T\in\mathcal{T}_h}\|v\|_{H^1(T)}^2\right)^{1/2} \leq Ch^{-1}\left(\sum_{T\in\mathcal{T}_h}\|v\|_{L^2(T)}^2\right)^{1/2} = Ch^{-1}\|v\|_{L^2(\Omega)}$$

for all $v \in V_h = \{v : v \text{ is measurable and } v|_T \in \mathcal{P}_T, \forall T \in \mathcal{T}_h\}$, where $\mathcal{P}_T \subseteq H^1(T) \cap L^2(T)$ and $h = \max_{T\in\mathcal{T}_h}\operatorname{diam}(T)$, and C depends on the polynomial degree and shape regularity of the mesh. This implies that

$$\|\nabla v\|_{L^2(\Omega)}^2 \leq C^2h^{-2}\|v\|_{L^2(\Omega)}^2.$$

Since $\mu_j^{k+1} \in V_{j,r} \subset V_j$ is a finite-dimensional space of piecewise polynomials on a shape-regular mesh $\mathcal{T}_{h,j}$, the standard inverse inequality (Brenner & Scott, Thm. 4.5.11) implies

$$\|\nabla\mu_j^{k+1}\|_{L^2(\Omega_j)} \leq C_jh_j^{-1}\|\mu_j^{k+1}\|_{L^2(\Omega_j)},$$

where C depends on the polynomial degree and the mesh regularity. This allows us to control the H^1-norm of μ_j^{k+1} in terms of its L^2-norm. Thus, the inequality becomes

$$\begin{aligned}\frac{d}{dt}\left\|\mu_j^{k+1}\right\|^2_{L^2(\Omega_j)} \leq \varepsilon \sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e^k_{l,r}\|^2_{H^1(\Omega_l)} + \tilde{\varepsilon}\sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|_{H^{1/2}(\Gamma_{jl})}+ \\ + \left(\frac{1}{\hat{\varepsilon}} + C_j^2h_j^{-2}\left(\sum_{l\in\mathcal{N}_j}\left(C_{1,l}C_{2,l}\frac{1}{\varepsilon} + C_{2,l}\frac{1}{\tilde{\varepsilon}}\right) - 2\right)\right)\|\mu_j^{k+1}\|^2_{L^2(\Omega_j)}+ \\ + \hat{\varepsilon}\|\Delta\eta_j^{k+1}\|^2_{L^2(\Omega_j)}.\end{aligned}$$

We can rewrite this as

$$\dot{\varphi}_j^{k+1}(t) \leq \beta_j(t)\varphi(t) + \alpha_j^k(t),$$

where

$$\begin{aligned}\varphi_j^{k+1}(t) &= \left\|\mu_j^{k+1}(t)\right\|^2_{L^2(\Omega_j)}, \\ \beta_j(t) &= \frac{1}{\hat{\varepsilon}} + C_j^2h_j^{-2}\left(\sum_{l\in\mathcal{N}_j}\left(C_{1,l}C_{2,l}\frac{1}{\varepsilon} + C_{2,l}\frac{1}{\tilde{\varepsilon}}\right) - 2\right), \\ \alpha_j^k(t) &= \varepsilon\sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e^k_{l,r}\|^2_{H^1(\Omega_l)} + \tilde{\varepsilon}\sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|^2_{H^{1/2}(\Gamma_{jl})} + \hat{\varepsilon}\|\Delta\eta_j^{k+1}\|^2_{L^2(\Omega_j)}.\end{aligned}$$

Then, by Grönwall's lemma, we have

$$\varphi_j^{k+1}(t) \leq \varphi_j^{k+1}(0)e^{\int_0^t\beta_j\,d\tau} + \int_0^t \alpha_j^k(s)e^{\int_s^t\beta_j\,d\tau}\,ds.$$

Since $\beta_j(t)$ is constant and $\mu_j^{k+1}(\cdot,0) = 0$ for all k, we have

$$\varphi_j^{k+1}(t) \leq \int_0^t \alpha_j^k(s)e^{\beta_j(t-s)}\,ds \leq \sup_{s\in[0,t]}\alpha_j^{k+1}(s)\cdot\int_0^t e^{\beta_j(t-s)}\,ds = \frac{e^{\beta_j t}-1}{\beta_j}\cdot\sup_{s\in[0,t]}\alpha_j^k(s) = M_j(t)\sup_{s\in[0,t]}\alpha_j^k(s).$$

Let us take $\varepsilon = \tilde{\varepsilon} = \hat{\varepsilon} = 1$, then

$$\begin{aligned}\varphi_j^{k+1}(t) &= \left\|\mu_j^{k+1}(t)\right\|^2_{L^2(\Omega_j)}, \\ \beta_j &= 1 + C_j^2h_j^{-2}\left(\sum_{l\in\mathcal{N}_j} C_{2,l}\left(C_{1,l}+1\right) - 2\right), \\ \alpha_j^k(t) &= \sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e^k_{l,r}\|^2_{H^1(\Omega_l)} + \sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|^2_{H^{1/2}(\Gamma_{jl})} + \|\Delta\eta_j^{k+1}\|^2_{L^2(\Omega_j)}.\end{aligned}$$

Then

$$\begin{aligned}\left\|\mu_j^{k+1}(t)\right\|^2_{L^2(\Omega_j)} \leq M_j(t)\Bigg(&\sup_{s\in[0,t]}\sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e^k_{l,r}\|^2_{H^1(\Omega_l)}+ \\ &+ \sup_{s\in[0,t]}\sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|^2_{H^{1/2}(\Gamma_{jl})} + \sup_{s\in[0,t]}\|\Delta\eta_j^{k+1}\|^2_{L^2(\Omega_j)}\Bigg).\end{aligned}$$

$$\sup_{s\in[0,t]}\alpha(s) = \sup_{s\in[0,t]}\sum_{l\in\mathcal{N}_j} C_{1,l}C_{2,l}\|e^k_{l,r}\|^2_{H^1(\Omega_l)} + \sup_{s\in[0,t]}\sum_{l\in\mathcal{N}_j} C_{2,l}\inf_{v_r\in V_{j,r}}\|\mathcal{T}_{jl}u_l^k - v_r\|^2_{H^{1/2}(\Gamma_{jl})} + \sup_{s\in[0,t]}\|\Delta\eta_j^{k+1}\|^2_{L^2(\Omega_j)}$$

$$\|e^k_{l,r}\|^2_{H^1(\Omega_l)} = \|\eta_l^k\|^2_{H^1(\Omega_l)} + \|\mu_l^k\|^2_{H^1(\Omega_l)} \leq \|\eta_l^k\|^2_{H^1(\Omega_l)} + C_l^2h_l^{-2}M_l(t)\sup_{s\in[0,t]}\alpha_l^k(s)$$